

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech (CL/EC/ ME) Sem. – I Examination 2009-10

ME-101 Engineering Graphics

Date: 18.12.2009, Friday

Time: 2:00 p.m. to 5:00 p.m.

Maximum Marks: 70

Instruction:

- 1) Attempt all the questions.
- 2) Figures to the right indicate full marks.
- 3) Make suitable assumptions if required.

SECTION I

Q 1 (a) The distance between a fixed point and a fixed line is 60 mm. Draw the locus of the moving point such that its distance from the fixed point is twice its distance from a fixed straight line. Name the curve. Also draw Tangent and Normal to the curve at any point P. 04

(b) A rectangular garden of area 196 m^2 is shown on a map by a similar rectangle of 4 cm^2 . Construct a plain scale to read up to a single meter and long enough to measure up to 9.8 decameters. Show the distances of (i) 2.2 decameter (ii) 9.7 decameter. 05

Q 2 Figure (1) shows the isometric view of a cross block. Draw using first angle projection method, the following views. (a) Front view in 'X' direction. (b) Right hand side view 10

OR

Q 2 Figure (2) shows the isometric view of a block. Draw using first angle projection method, the following views. (a) Front view in 'X' direction. (b) Sectional Left hand side view along section AA. 10

Q 3 (a) Draw the projection of a line AB, 80 mm long which end A is 16 mm below the HP & 20 mm behind the VP and the end B is in the first quadrant. The line is inclined at 45° to the HP & 30° to the VP. 08

(b) ABCD is a rhombus of diagonals $AC = 100 \text{ mm}$ and $BD = 70 \text{ mm}$. Its corner A is in the H.P. and the plane is inclined to the H.P. such that the plan appears to be a Square. The plan of the diagonal AC makes an angle of 20° to V.P. Draw the projections of the plane and find its inclination with H.P. 08

OR

Q3 (a) A and B are two points in space. The point A is in V.P., and 24 mm above H.P. The point B is in H.P. and 36 mm from V.P. The point A is 54 mm away from B. Draw the top view and front view of the straight line AB and also determine its inclinations with H.P. & V.P. 08

- (b) A thin 45° set square with longest side measuring 50 mm rests on VP on this side such that the surface makes an angle of 30° with VP. Draw the projections of Set Square when the side resting on VP is inclined at 40° to HP. 08

SECTION II

- Q 4 (a) Draw the isometric view of Figure :3 08
 (b) Give the application of Auto CAD and Pro/E 02
 (c) Explain the system of Dimensioning 02

- Q 5 (a) A tetrahedron of 30 mm side is resting with one of its edges on H.P. The edge on which it rests is inclined at 45° to V.P, and a face containing that edge is inclined 30° to H.P. Draw the projections of solid. 07

- (b) A pentagonal pyramid, side of base 20 mm and height 60 mm is resting on HP on its base with a side of base parallel to and nearer to VP. A sectional plane AVP perpendicular to HP and inclined to VP by 45° cuts the pyramid at a distance 10 mm away from the axis. Draw the plan, sectional elevation and true shape of the section. 08

OR

- Q 5 (a) A pentagonal pyramid of side of base 30mm and height 55 mm is lying on the H.P. on one of its side such that corner opposite to that side is 15 mm above the H.P. Draw the projections of the solid and also project elevation on an auxiliary plane when the side on which it is resting is parallel to V.P 07

- (b) A cone of base diameter 50 mm and height 70 mm is resting on HP on one of its generator with axis parallel to VP. It is cut by horizontal section plane passing through a point on the axis 45 mm away from the apex. Draw elevation and sectional plan. 08

- Q 6 A square pyramidal frustum has its bottom side 60mm, topside 30mm and height 70mm. It's resting on H.P. on its bottom keeping two side of the frustum parallel to V.P. Draw development of the frustum showing in it the string connecting the mid point of the side of one face of top with the mid point of bottom of the opposite face by shortest length. Show string in the projections of the frustum. 08

OR

- Q 6 A pentagonal prism, side of 28mm and height 65mm has an edge of its base on the H.P. and the axis parallel to V.P. An A.I.P at 60 degree cuts the prism and passes through the point on the axis 50 mm above the base. Draw elevation, sectional plan and also draw true shape of the section , development of remaining part of solid. 08

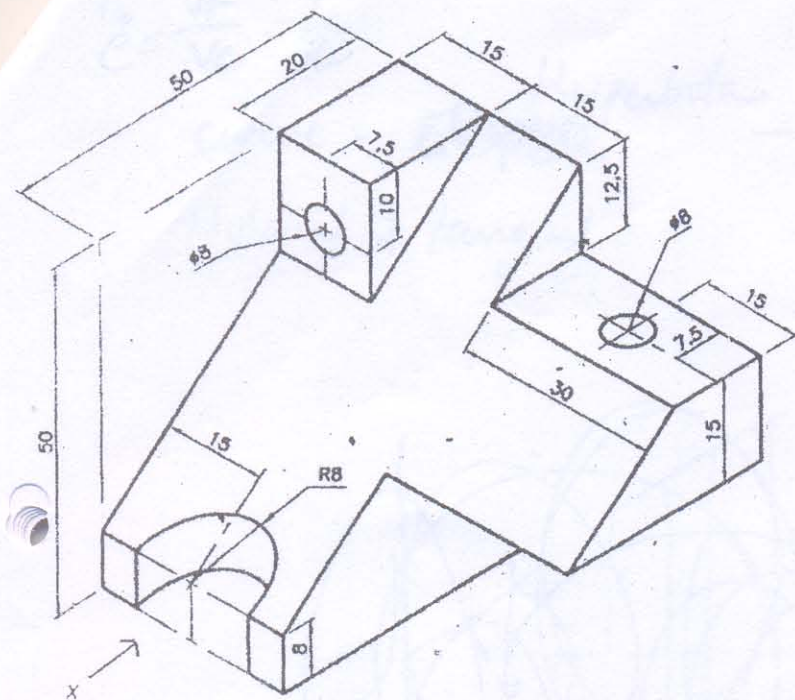


Fig. 1

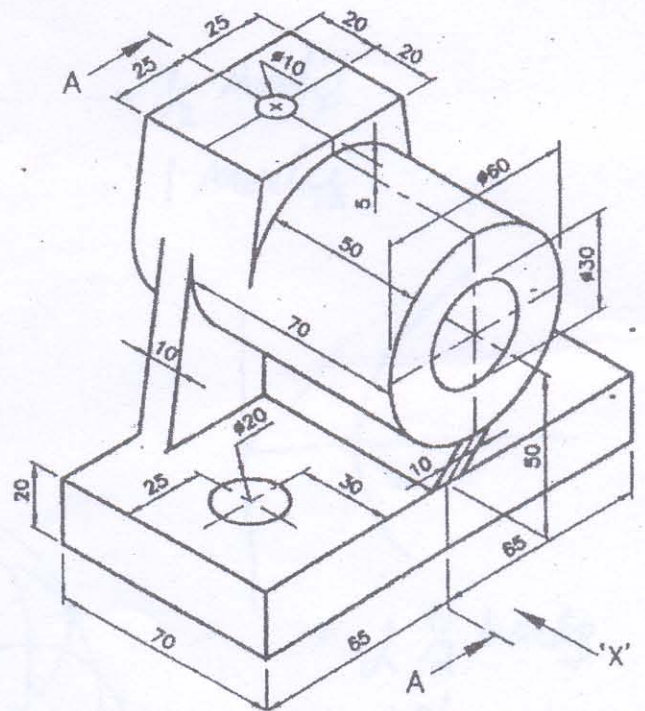


Fig. 2

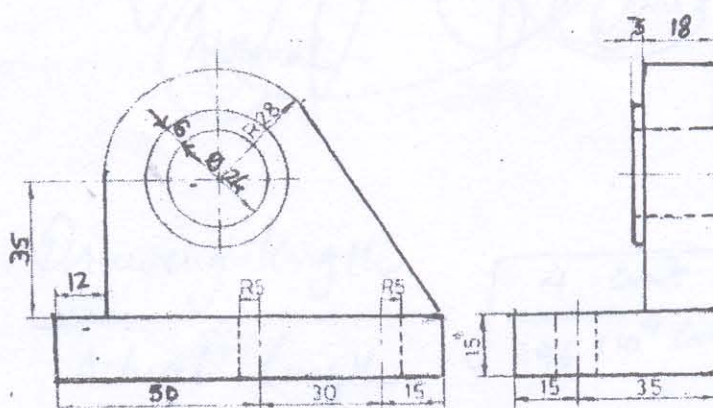


Fig. 3